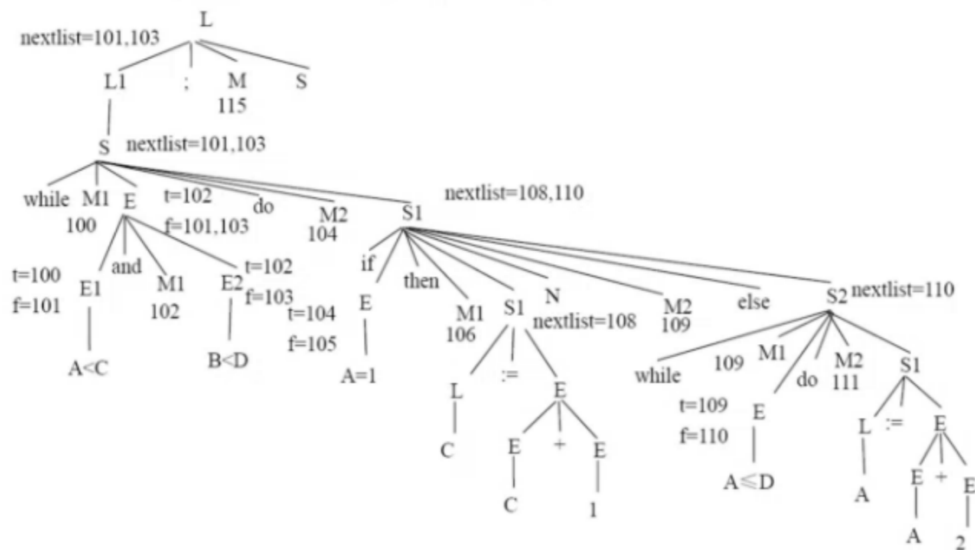


7. While A<C and B<D do

if A=1 then C:=C+1 else

while A≤D do A:=A+2;



100: (j<, A, C, 102)
 101: (j, -, -, 115)
 102: (j<, B, D, 104)
 103: (j, -, -, 115)
 104: (j=, A, '1', 106)
 105: (j, -, -, 109)
 106: (+, C, '1', T1)
 107: (: =, T1, -, C)
 108: (j, -, -, 100)
 109: (j≤, A, D, 111)
 110: (j, -, -, 100)
 111: (+, A, '2', T2)
 112: (: =, T2, -, A)
 113: (j, -, -, 109)
 114: (j, -, -, 100)
 115:

五、按照下述给出的翻译模式，写出布尔式 A or (B and c<d or not F)的四元式序列。(15 分)

产生式	语义规则
$E \rightarrow E1 \text{ or } M E2$	{ backpatch(E1.falselist, M.quad); E.truelist:=merge(E1.truelist, E2.truelist); E.falselist:=E2.falselist }
$E \rightarrow E1 \text{ and } M E2$	{ backpatch(E1.truelist, M.quad); E.truelist:=E2.truelist; E.falselist:=merge(E1.falselist,E2.falselist) }
$E \rightarrow \text{not } E1$	{ E.truelist:=E1.falselist; E.falselist:=E1.truelist }
$E \rightarrow (E1)$	{ E.truelist:=E1.truelist; E.falselist:=E1.falselist }
$E \rightarrow \text{id1 rel op id2}$	{ E.truelist:=makelist(nextquad); E.falselist:=makelist(nextquad+1); emit('j' rel op.op ' , id 1.place ' , id 2.place ' , ' 0'); emit('j, -, -, 0') }
$E \rightarrow \text{id}$	{ E.truelist:=makelist(nextquad); E.falselist:=makelist(nextquad+1); emit('jnz' ' , id .place ' , ' - ' , ' 0'); emit(' j, -, -, 0') }
$M \rightarrow \epsilon$	{ M.quad:=nextquad }